Assignment: Product Inventory Analysis for Apex Global Warehouse

Overview

In this assignment, you will apply your SQL and data analysis skills to a real-world dataset. You will act as a Data Analyst for a large warehouse business, using BigQuery to answer critical business questions that will drive the company's sales strategy.

Part 1: Instructions & Environment Setup

Before you begin, you must set up your BigQuery environment. This assignment is designed to be Sandbox friendly, meaning you can complete it without a credit card or incurring charges.

1. **Account Setup:** Follow the instructions in your GDA course under the section "Set up your BigQuery account for more information".
2. **Data Upload:**
   - Download the amazon.csv file.
   - Follow the steps in the GDA section "Optional: Upload the store transactions dataset to BigQuery" to upload this CSV.
3. *****Important Table Details***:**
   - Name your dataset: warehouse_analysis
   - Name your table: amazon_products
   - Ensure you check the "Auto-detect Schema" box during upload.
     - Your Schema should look like this

Part 2: The Business Story

The Scenario: You have just been hired as a Junior Data Analyst at Apex Global Warehouse, a major distribution hub for top-tier electronics and home goods. The company is preparing for its Quarterly Inventory Review, and the CEO needs to know which product lines are driving the most value and which ones have stalled.

- While the finance team has already cleaned the currency symbols for you, the data still has "traps" that could ruin your report:
- Corrupted Ratings: Some customer reviews didn't sync correctly, resulting in strange characters like | instead of a number.
- Complex Categories: Products are grouped into long strings (e.g., Electronics|HomeAudio|Accessories). You'll need to figure out how to find specific keywords within those strings.
- Massive Volume: With thousands of rows, you can't just scroll through the data, instead, you must use SQL to summarize the big picture.

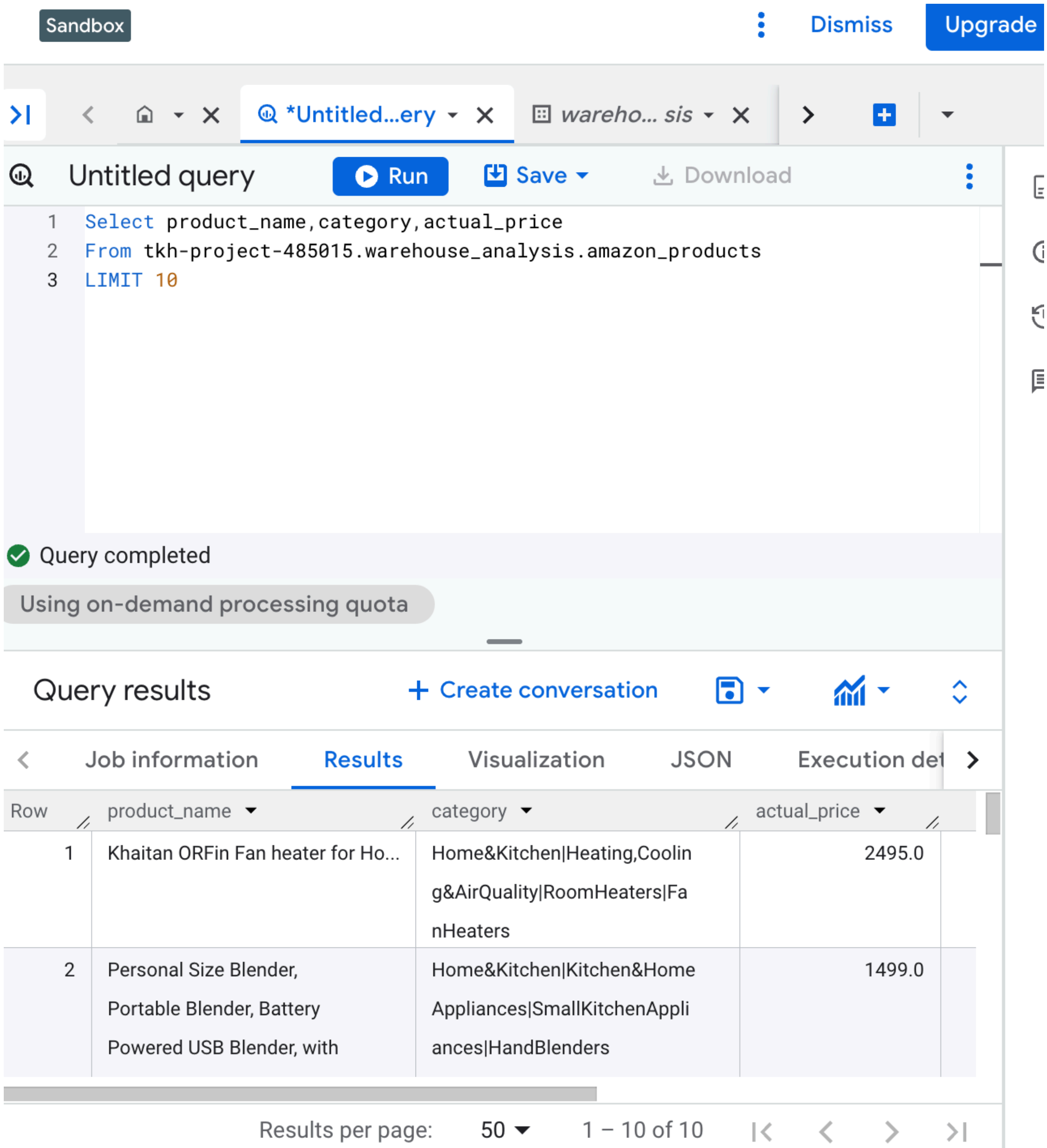
Your Mission: Use BigQuery to navigate these issues and extract 10 key insights. Your final report will help the CEO decide which "Premium" items to promote and which "Budget" items to clear out of the warehouse.

Part 3: Business Questions & SQL Answers

1. Quick Stock Check

Question: Management wants a quick look at our inventory. Show the *product_name*, *category*, and *actual_price* for the first 10 products.

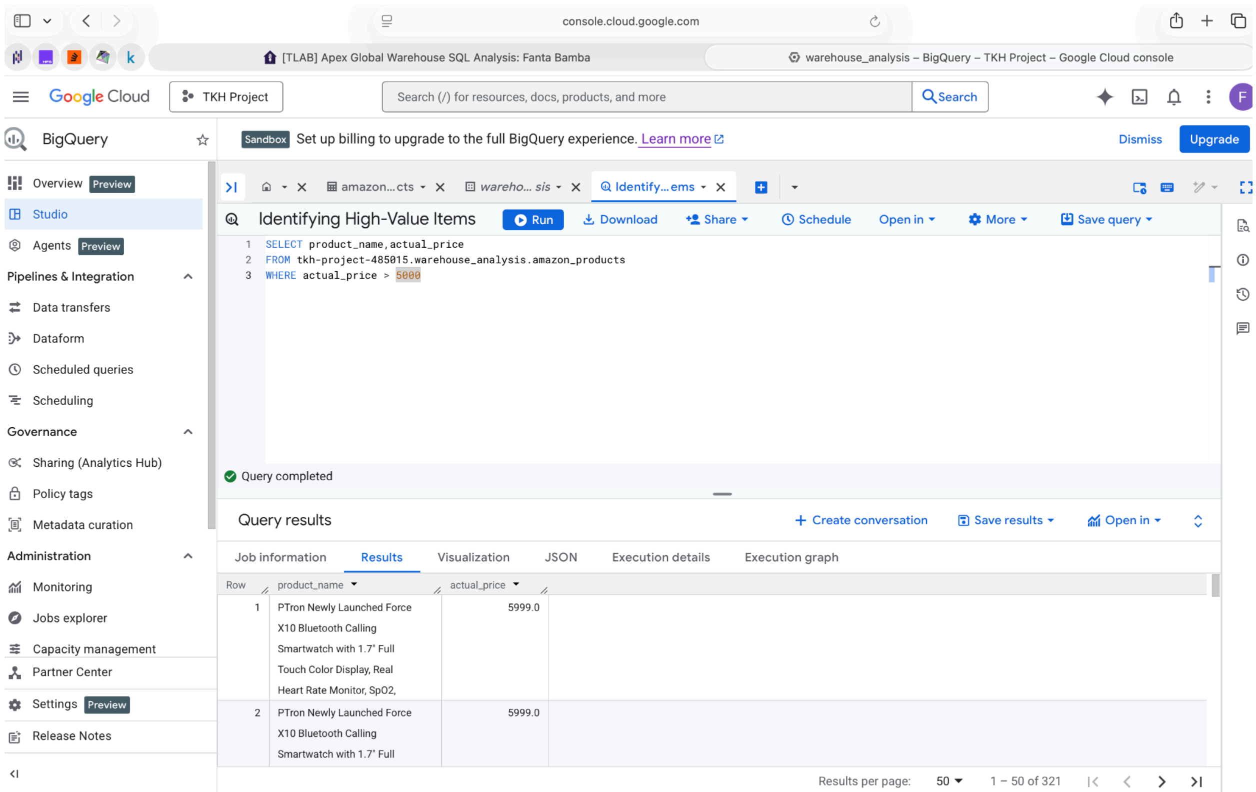
Tip: Think about how to "cut off" the results so you don't see the entire warehouse list at once.



2. Identifying High-Value Items

Question: Find all products where the *actual_price* is greater than 5,000. Show the name and the price.

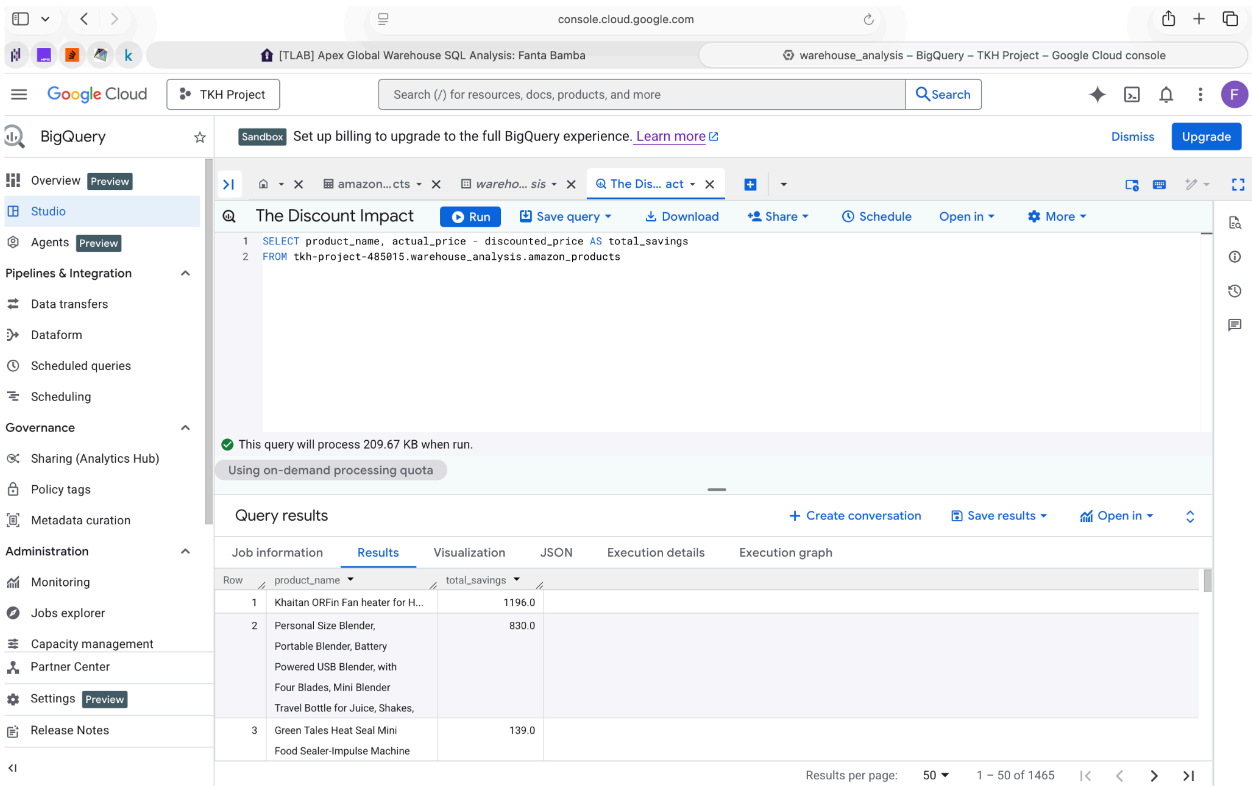
Tip: Use a filter to only look for items that meet a specific financial threshold.



3. The Discount Impact

Question: Calculate the total amount saved (Actual Price - Discounted Price) for each product. Call this new column *total_savings*. (is this AS)

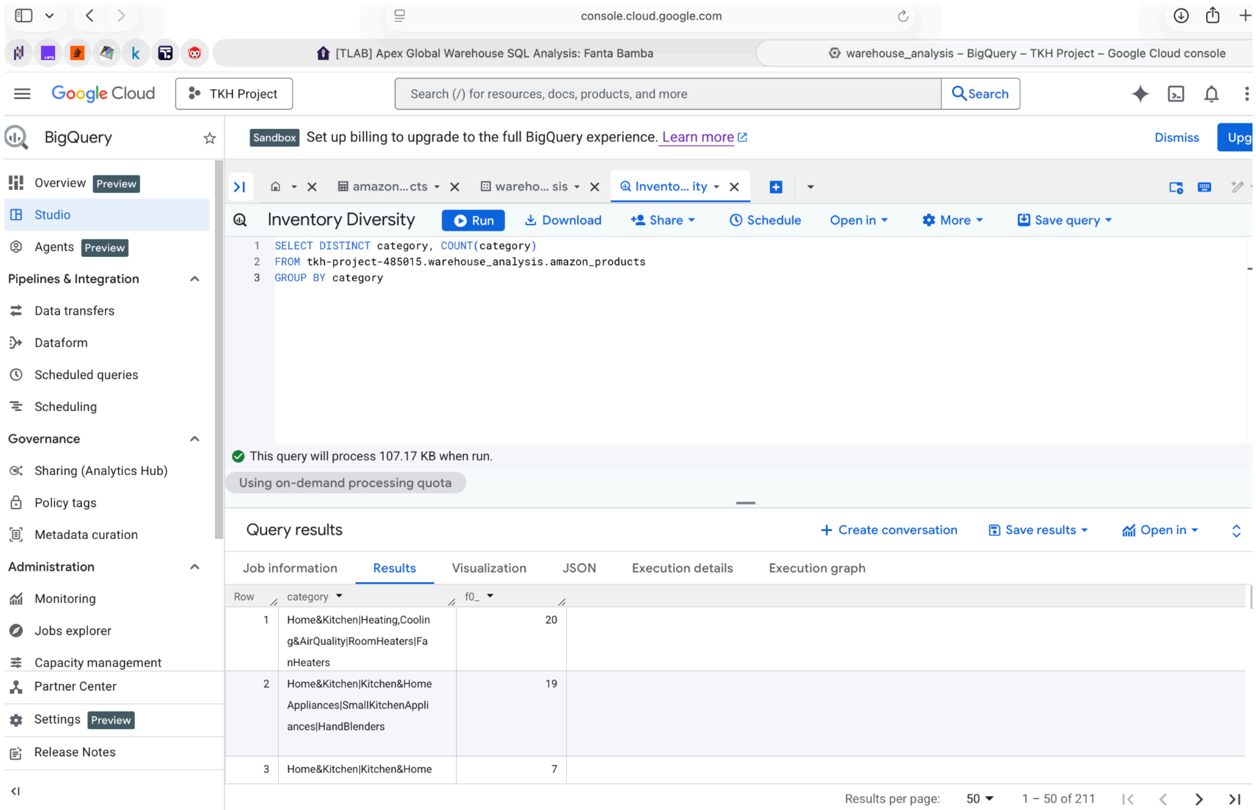
Tip: You can perform basic subtraction between columns and then organize the results to see the biggest savers first.



4. Inventory Diversity

Question: How many unique products do we have in each category? Show the category and the count.

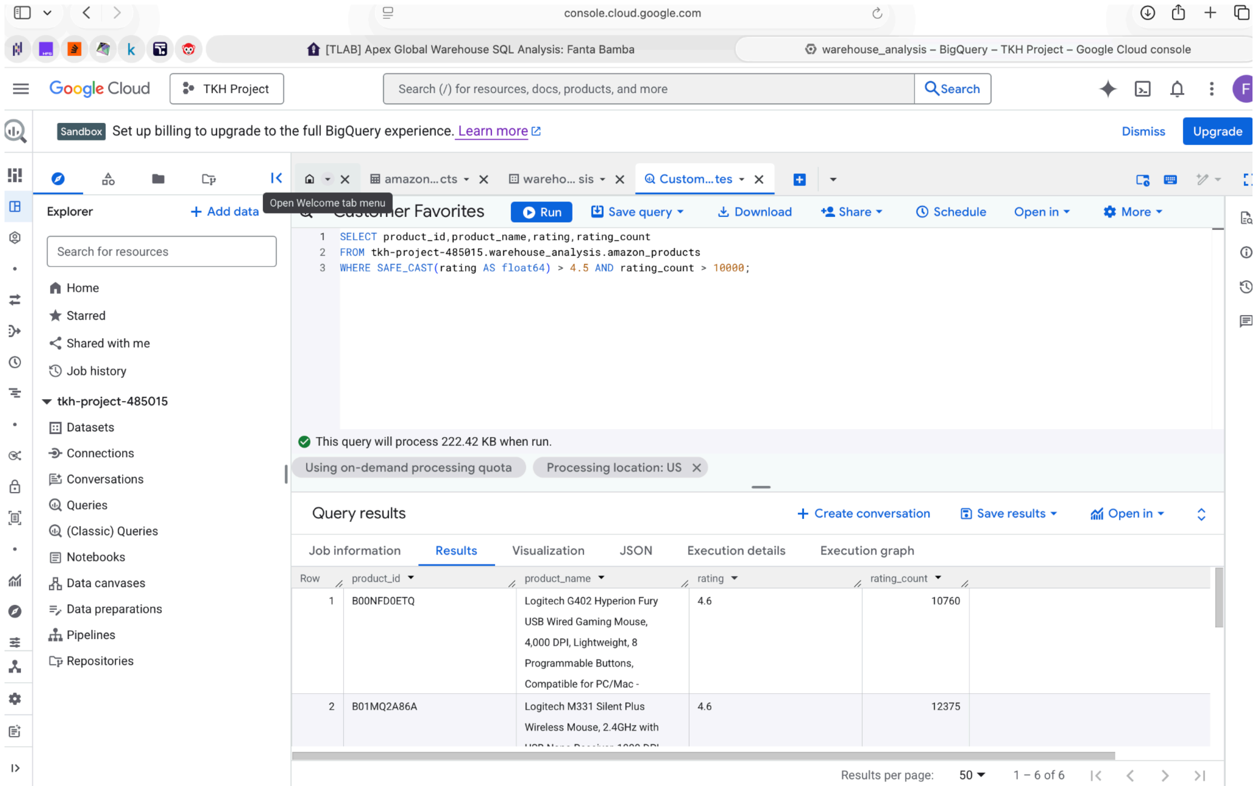
Tip: You'll need to "bucket" your data by category and then use a function to count what is inside each bucket.



5. Customer Favorites

Question: We need a list of highly-rated products. Find products that have a *rating* higher than 4.5 and more than 10,000 *rating_count*.

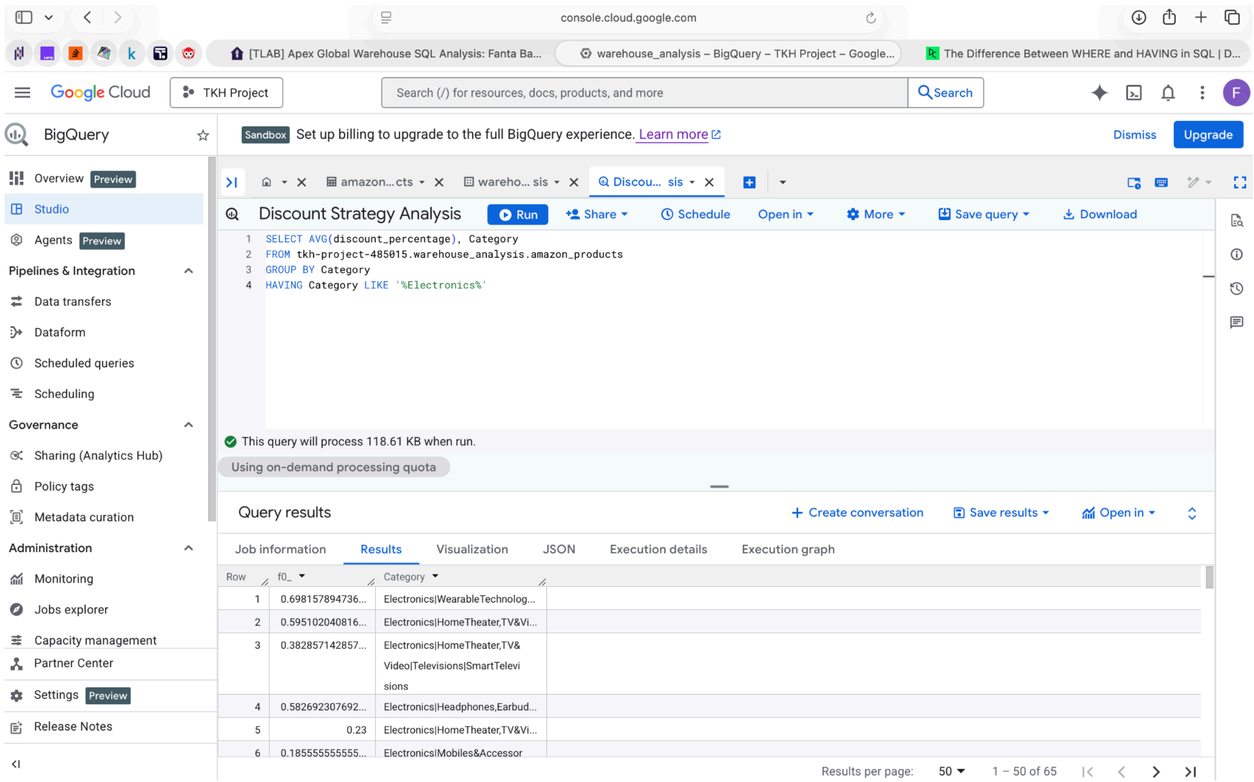
Tip: Look for a way to combine two different conditions so the product must meet both requirements.



6. Discount Strategy Analysis

Question: What is the average *discount_percentage* for products in the 'Electronics' category?

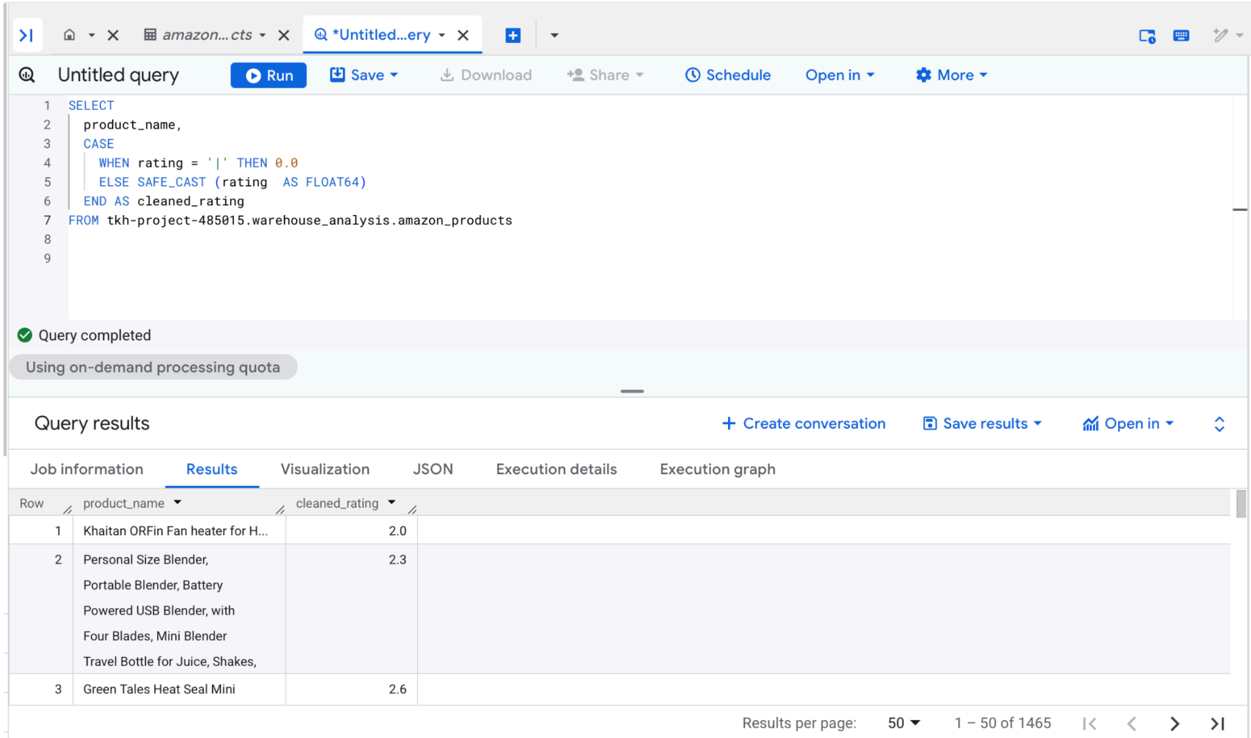
Tip: Use a math function to find the middle value of the discounts, and use a wildcard to find any category that contains the word "Electronics."



7. Cleaning Up the Ratings

Question: Some ratings are missing or corrupted (like the '|' character). Use a CASE statement to show the *product_name* and a new column that turns any rating of '|' into 0.0, otherwise keep the original rating.

Tip: Create a logic "switch" that checks the value of the rating and changes it only if it matches a specific "bad" character.

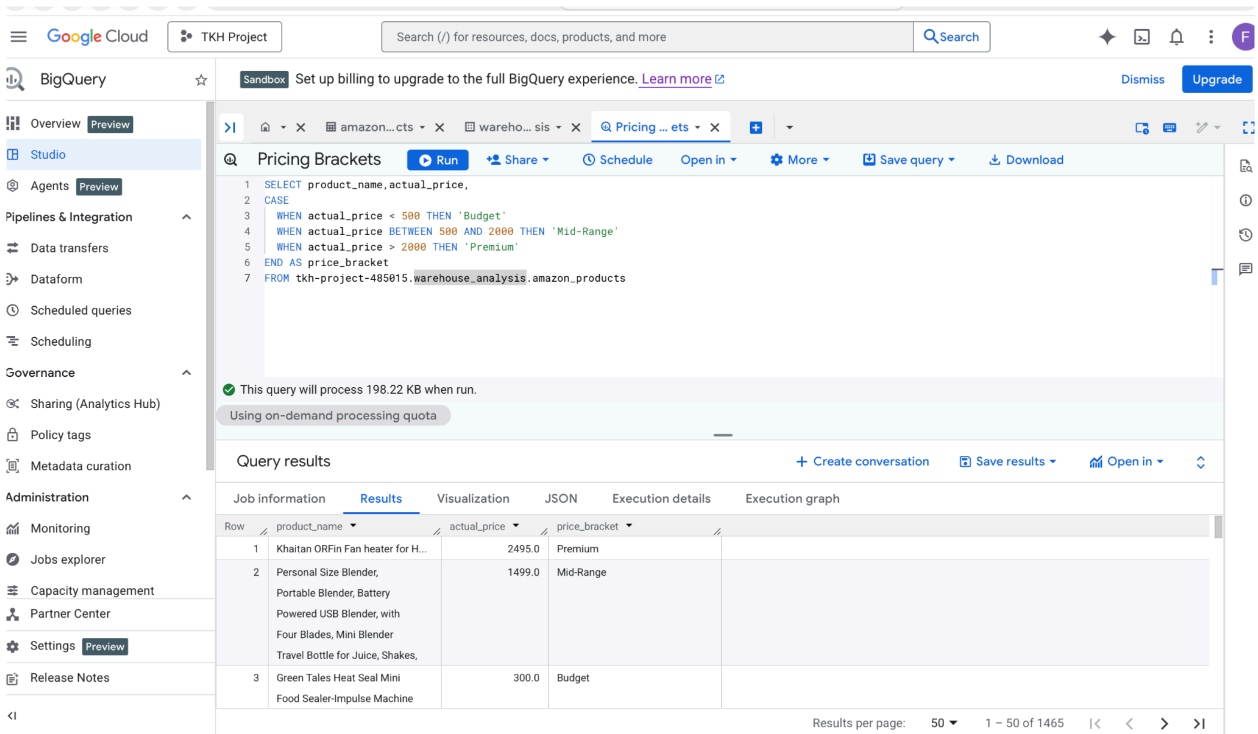


8. Pricing Brackets

Question: Create a column called *price_bracket* where:

- actual_price < 500 is 'Budget'
- actual_price between 500 and 2000 is 'Mid-Range'
- actual_price > 2000 is 'Premium'

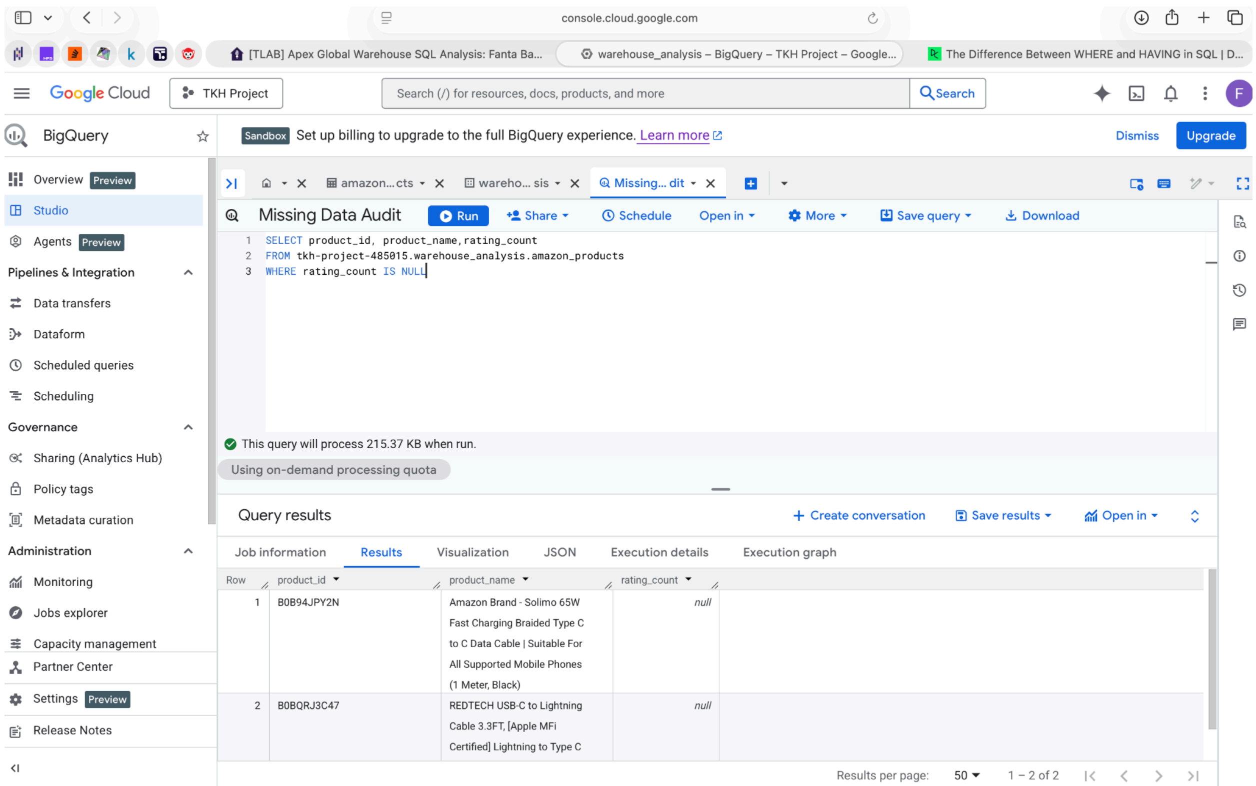
Tip: Use multiple logic branches to categorize your products based on their cost ranges. Once do the work AS



9. Missing Data Audit

Question: Identify any products where the *rating_count* is missing. This is crucial for our data integrity report.

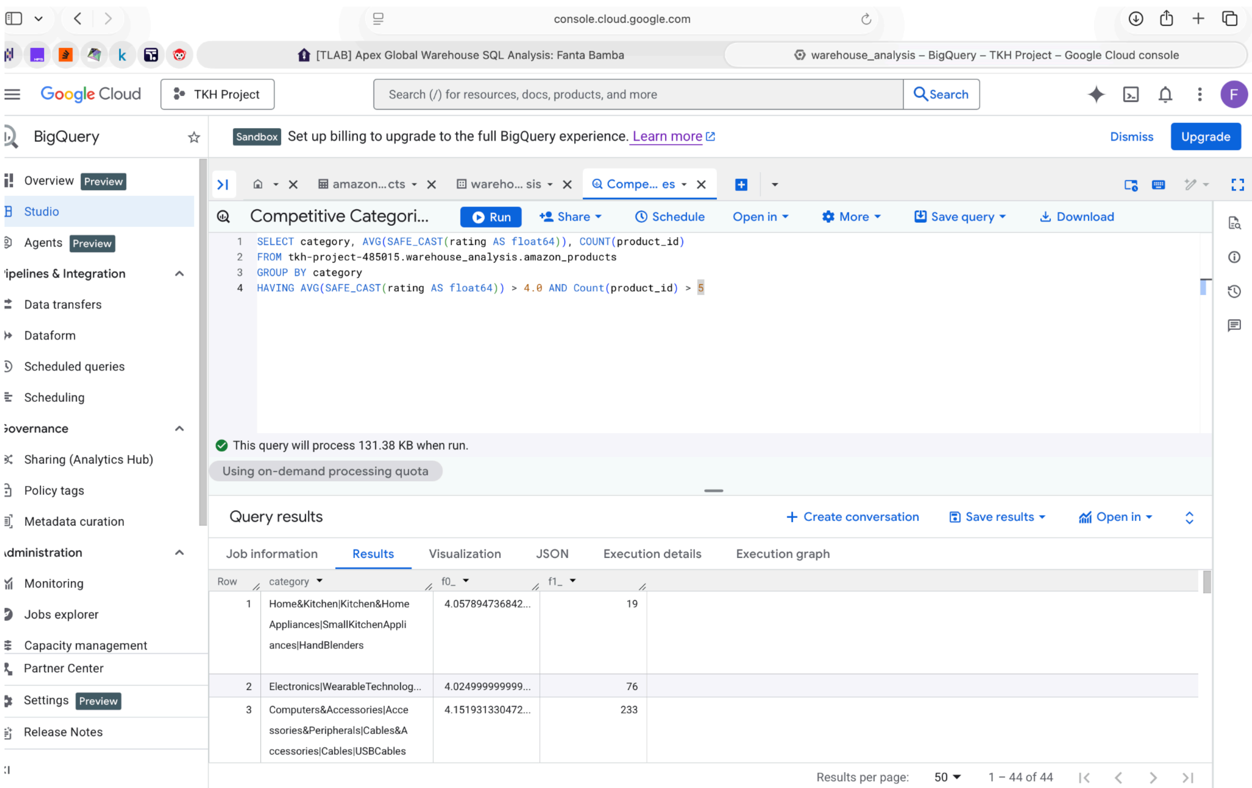
Tip: Search for a specific state of data where a cell is completely empty. NULL



10. Competitive Categories

Question: Which categories have an average rating above 4.0? Only show categories that have more than 5 products.

Tip: After you group your data, you'll need to filter the results of your math rather than the raw rows.



List of Necessary Queries to know:

- SELECT ... FROM: The foundation of every query.
- AS (Aliasing): Giving your calculated columns a professional name.
- WHERE vs HAVING: Understanding when to filter raw data and when to filter grouped data.
- GROUP BY: The key to summarizing data by category.
- ORDER BY DESC: How to sort your best-selling or most-expensive items to the top.
- CASE WHEN: How to create "if-then" logic inside your SQL tables.
- LIKE '%Text%': Using wildcards to find keywords within long category strings.
- IS NULL: The standard way to find missing information.